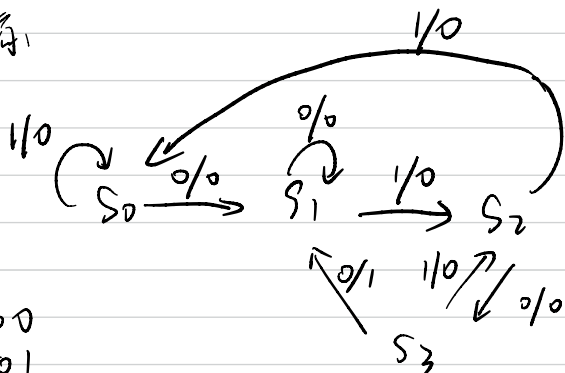


1. 用JK触发器设计一个检测输入序列0100的同步时序电路
 输入数据可重复使用. 例如: 串行输入 00100100

输出 00001001

JK触发器: $Q^{n+1} = J\bar{Q} + KQ$

建立状态图并赋值:



分配状态:

S ₀	00
S ₁	01
S ₂	10
S ₃	11

0101
=

状态	$Q_1^n Q_0^n$	输入X	次态 $Q_1^{n+1} Q_0^{n+1}$	输出Y
	00	0	01	0
	00	1	00	0
	01	0	01	0
	01	1	10 ✓	0
	10	0	11 ✓	0
	10	1	00	0
	11	0	01	1
	11	1	10 ✓	0

输出方程

~~$Q_1^n Q_0^n$~~

00 01 11 10

0
1

$$Y = \bar{X} Q_1^n Q_0^n$$

Q_0^{n+1} 卡诺图:

$X \backslash Q_1^n Q_0^n$	00	01	11	10
0	1	1	1	1
1				

$$Q_0^{n+1} = \bar{X} (Q_0^n + \bar{Q}_0^n)$$

Q_1^{n+1} 卡诺图:

$X \backslash Q_1^n Q_0^n$	00	01	11	10
0				1
1		1	1	

$$Q_1^{n+1} = Q_1^n (X Q_0^n + \bar{X} \bar{Q}_0^n) + X \bar{Q}_1^n Q_0^n$$

$$Q_1^{n+1} = X Q_0^n + \bar{X} Q_1^n \bar{Q}_0^n$$

利用 $Q^{n+1} = J \bar{Q}^n + \bar{K} Q^n$ 得驱动方程:

$$\left. \begin{aligned} J_0 &= \bar{X} & K_0 &= X \\ J_1 &= X Q_0^n & \bar{K}_1 &= \bar{X} \bar{Q}_0^n \end{aligned} \right\}$$

$$K_1 = X + Q_0^n$$

检查是否自启, 由于没有无效状态, 故可以.

2. 用 D 触发器设计一个同步六进制计数器:

计数序列 $000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 101 \rightarrow 000$

D 触发器满足 $Q^{n+1} = D$

现态 ($Q_2^n Q_1^n Q_0^n$)	次态 ($Q_2^{n+1} Q_1^{n+1} Q_0^{n+1}$)	输出 Z
000	001 ✓	0
001	010	0
010	011 ✓	0
011	100	0
100	101 ✓	0
101	000	1
110	X (无效)	
111	X	

建卡诺图, $D_2 = Q_2^{n+1}$

$Q_2^n Q_1^n Q_0^n$	00	01	11	10
0			1	
1		D	X	X

$$D_2 = Q_1^n Q_0^n + Q_2^n \bar{Q}_0^n$$

$$D_1 = Q_1^{n+1}$$

$Q_2^n Q_1^n Q_0^n$	00	01	11	10
0			1	
1			X	X

$$D_1 = \bar{Q}_2^n \bar{Q}_1^n Q_0^n + Q_1^n \bar{Q}_0^n = \bar{Q}_2^n (Q_1^n \oplus Q_0^n)$$

$$D_0 = Q_0^{n+1}$$

$$\begin{array}{r} Q_2 Q_1 Q_0^n \\ \hline Q_2 \end{array}$$

00 01 11 10

0	1		1
1	1	x	x

$$D_0 = \overline{Q_0^n}$$

检查能否自启动:

110 代入驱动方程 $D_2 D_1 D_0 = 101$

进入有效循环

111 代入得 $D_2 D_1 D_0 = 100$

可以自启动